



1. Introduction

Walmart operates one of the world's largest retail supply chains, serving millions of customers every week. To ensure product availability while controlling costs, Walmart must decide how many units of each product to order before the actual customer demand is known. Demand is uncertain, but forecasts provide several possible demand scenarios with associated probabilities. This paper develops and solves a stochastic inventory optimization model to determine the optimal order quantities for five consumer products.

2. Problem Statement

Walmart must decide how many units to order of five consumer products for next week: laundry detergent, paper towels, bottled water, breakfast cereal, and baby diapers. The orders are placed before demand is known. If Walmart orders too many units, the excess inventory incurs holding costs (warehouse space, handling, etc.). If Walmart orders too few units, unsatisfied demand incurs shortage costs (lost sales, customer dissatisfaction, and emergency replenishment). Walmart has a limited warehouse capacity of 16,000 units.

Objective: Determine the order quantities for the five products that minimize the expected total holding and shortage cost, subject to the warehouse capacity constraint.

3. Data

3.1 Products

Product	Symbol	Description
Laundry Detergent	D	Keep clothes clean
Paper Towels	P	For everyday use
Bottled Water	W	Stay hydrated
Breakfast Cereal	C	Popular breakfast
Baby Diapers	B	Essential for families

3.2 Demand Forecast and Scenario Probabilities

Product	Scenario 1 Very Low	Scenario 2 Low	Scenario 3 Normal	Scenario 4 High	Scenario 5 Very High
Detergent (D)	1,854	2,163	2,575	2,987	3,502
Paper Towels (P)	2,266	2,678	3,193	3,811	4,532
Bottled Water (W)	3,090	3,914	4,841	5,974	7,210
Cereal (C)	1,648	1,957	2,369	2,833	3,399
Diapers (B)	1,442	1,751	2,163	2,575	3,090

Scenario Probabilities					
Scenario (s)	1	2	3	4	5
Probability (p_s)	0.10	0.20	0.40	0.20	0.10

Note: Scenario 3 (Normal demand) is the most likely, with 40% probability.

3.3 Unit Costs

Product	Holding Cost h_i (\$/unit)	Shortage Cost c_i (\$/unit)
Detergent (D)	1.50	8
Paper Towels (P)	0.80	6
Bottled Water (W)	0.50	5
Cereal (C)	1.20	7
Diapers (B)	1.00	12

3.4 Warehouse Capacity

Total units that can be ordered:

$$K = 16,000 \text{ units}$$

4. Mathematical Formulation

4.1 Notation

Sets / Indices	
i	Product index, $i \in \{D, P, W, C, B\}$
s	Demand scenario index, $s \in \{1, 2, 3, 4, 5\}$
Parameters	
d_{is}	Forecasted demand for product i in scenario s (units)
p_s	Probability of scenario s , $\sum_s p_s = 1$
h_i	Holding cost per unit of product i (\$/unit)
c_i	Shortage cost per unit of product i (\$/unit)
K	Warehouse capacity (units), $K = 16,000$
Decision Variables	
q_i	Order quantity for product i (units, first-stage)
I_{is}	Ending inventory of product i in scenario s (units)
U_{is}	Unmet demand of product i in scenario s (units)

4.2 Model

Objective Function

Minimize the expected total holding and shortage cost:

$$\min Z = \sum_{s \in S} p_s \sum_{i \in I} (h_i I_{is} + c_i U_{is}) \quad (1)$$

Subject to

1. Inventory balance (for all $i \in I, s \in S$):

$$q_i + U_{is} - I_{is} = d_{is} \quad (2)$$

2. Warehouse capacity:

$$\sum_{i \in I} q_i \leq K \quad (3)$$

3. Nonnegativity and integrality:

$$q_i \in \mathbb{Z}_+, \quad I_{is} \geq 0, \quad U_{is} \geq 0 \quad (4)$$

5. Solution Results (Using Gurobi)

The model was implemented in Python using Gurobi and solved to optimality. The results are shown below.

Optimal Order Quantities		Minimum Expected Total Cost
Product	Order Quantity (units)	\$ 7,847.76
Detergent (D)	2,987	
Paper Towels (P)	3,193	
Bottled Water (W)	4,841	
Cereal (C)	2,404	
Diapers (B)	2,575	
Total	16,000	

- ✓ Uses full warehouse capacity
- ✓ Balances holding and shortage costs
- ✓ Prioritizes high-penalty items (e.g., diapers)
- ✓ Minimizes expected total cost
- ✓ Helps keep shelves stocked and customers satisfied

6. Conclusion

This study develops a two-stage stochastic inventory optimization model for Walmart using forecasted demand scenarios. The model determines the optimal order quantities for five consumer products under a warehouse capacity constraint. The Gurobi solution recommends ordering 16,000 units in total, with product quantities that minimize the expected holding and shortage costs. The approach illustrates how data-driven optimization can improve inventory decisions, reduce costs, and enhance customer service in a real-world retail supply chain.

Key Takeaway: Using probabilistic forecasts and optimization, Walmart can make smarter inventory decisions today and build a more resilient, customer-focused supply chain for tomorrow.